

CSI 3140 — WWW Structures, Techniques and Standards

Laboratory 1 — Web Foundations, HTML5 Structure, Links, Forms, and Accessibility-Aware Markup

Professor: Mohamed Ali Ibrahim, PhD, PEng **Term:** Spring/Summer 2026

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1. Website Description

Our website is a fictional fitness club called **Uranus Fitness**; a child company of Planet Fitness offering different plans to sign up. The site introduces the club, describes its programs and workshop structure, presents a membership-plan schedule with pricing in a structured table, and lets visitors register for a membership tier through an accessible HTML form. The build employs standards-based HTML5 and accessibility-aware markup; CSS is used.

2. File Structure

Lab1/

— index.html	– Home page introducing Uranus Fitness (hero image + nav)
— about.html	– Club description: headings, lists, figure, external + email links
— schedule.html	– Membership-plan schedule as a captioned table
— registration.html	– Accessible registration form
— styles.css	– Basic CSS for layout, banner, readability
— img/	
— logo.png	– Banner logo, used in header on all pages
— old_guypng.png	– Hero image on the home page

3. Part A — Web Communication Process

When a browser loads one of our pages:

1. **Browser as client.** The browser (Chrome, Firefox) acts as the client, initiating a request on the user's behalf.
2. **URL.** The user enters/clicks a URL such as `https://localhost/index.html`, identifying the protocol (https), the server's domain name, and the path to the resource.
3. **DNS.** The browser contacts a DNS server to resolve the domain name into the IP address of the actual web-server machine.
4. **HTTP request.** Using that IP, the browser sends an HTTP get request for the resource (ex. `index.html`).
5. **Web server.** The server locates the file and returns an HTTP response with the content and a status code.
6. **HTTP response.** The browser parses the HTML, requests linked resources (CSS, images), and renders the page.
7. **Local vs. served files.** During development we open files via a `file://` path, so no DNS/HTTP occurs — the browser reads the file directly. When deployed, the full request/response cycle applies. Local development also requires relative paths to be correct from each file's location.

4. Part B — HTML5 Semantic Structure

Every page follows the standard HTML5 skeleton:

```
html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Page Title</title>
    <link rel="stylesheet" href="styles.css">
  </head>
  <body>
    <header> ... logo + <nav> ... </header>
    <main>
      <section> ... </section>
    </main>
    <footer> ... copyright ... </footer>
  </body>
</html>
```

Semantic elements used across the site:

Element

Purpose

<code><header></code>	Site banner (logo) and primary navigation on every page
<code><nav aria-label="Main navigation"></code>	Marks the primary nav block; aria-label describes it for screen readers
<code><main></code>	Wraps the unique primary content of each page
<code><section></code>	Groups thematically related content
<code><footer></code>	Copyright on every page
<code><h1>–<h3></code>	Logical heading hierarchy
<code> / </code>	Unordered and ordered lists
<code><table>, <caption>, <thead>, <tbody>, <th></code>	Membership-plans data table
<code><form>, <fieldset>, <legend></code>	Grouped, accessible form controls
<code><figure> / <figcaption></code>	Image with caption (about.html)

5. Part C — Links, Images, Lists, and Tables

Navigation and links

- Internal nav links on every page (relative paths) to index.html, about.html, schedule.html, registration.html.
 - Home index page can be clicked in the nav bar with the logo/image “Uranus Fitness”
- External link: about.html → Uranus Fitness YouTube channel (target="_blank" rel="noopener").
- Email link: about.html → `<a href="mailto:info@uranusfitness.example">`.
- Internal links: the "Select" links in the schedule table point to registration.html.

Images

- `img/logo.png` - header banner on all pages; alt="Uranus Fitness logo".
- `img/hero.png` - hero image on index.html; descriptive alt text.

Lists

- about.html: an unordered list of club offerings and an ordered list of the workshop session structure.
- schedule.html: an unordered list of program outcomes.

Table

- schedule.html: <caption>Uranus Fitness Plans</caption>, a <thead> of <th> headers (Plan, Duration, Type of Workout, Price, Select), and a <tbody> with three rows.

6. Part D — Accessible HTML Form

registration.html is grouped into <fieldset> blocks with <legend> labels and meets the six-control minimum:

Control	Type	Validation
Full name	<input type="text">	required, minlength="2"
Email	<input type="email">	required
Age group	<select>	required
Membership tier	<input type="radio"> group	required
About yourself	<textarea>	maxlength (optional)
Submit	<button type="submit">	

Accessibility features

- Every control has an explicit <label> linked via matching for/id
- Related controls grouped with <fieldset>/<legend>.
- HTML5 validation (required, type="email", minlength)
- Fully keyboard-navigable: Tab between fields, Enter to submit.

7. Part E — Accessibility and Validation Checklist

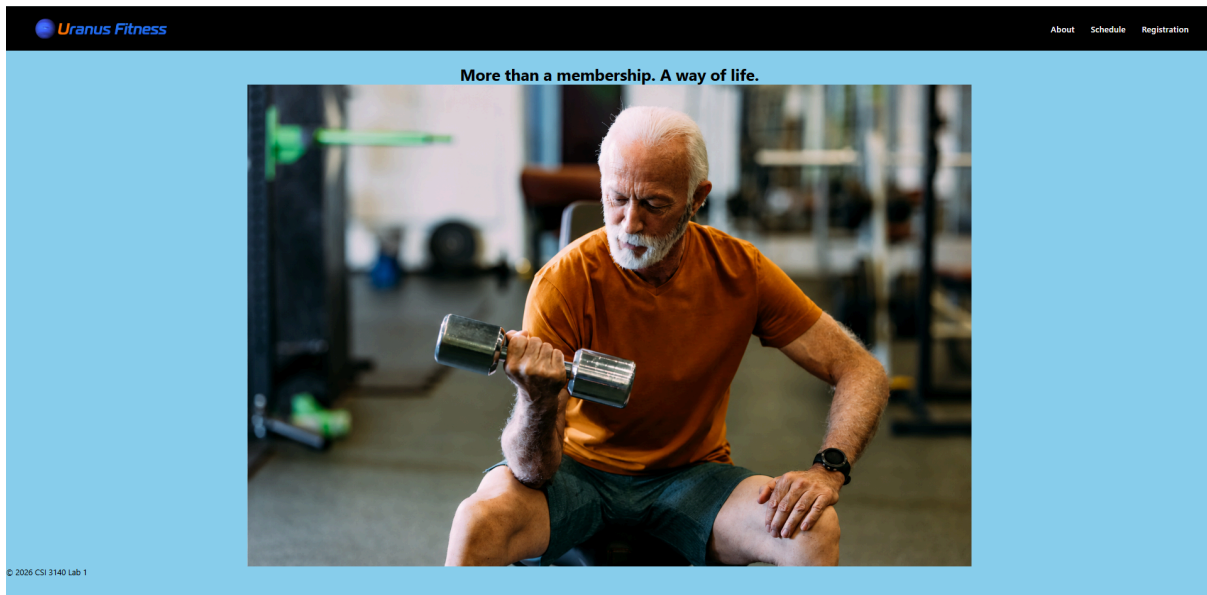
Item	Completed?
Each page has exactly one main h1 heading	yes
Headings follow a logical order without skipping levels	yes
Navigation links use meaningful text	Yes
Images have meaningful alt text (empty alt only when decorative)	Yes
Tables include captions and header cells	Yes
Form controls have explicit labels	Yes
The form can be navigated using the keyboard	Yes
The HTML files were checked using a validator	Yes — nuHTML
Major validation errors were fixed	yes

The pages were tested in at least one modern browser

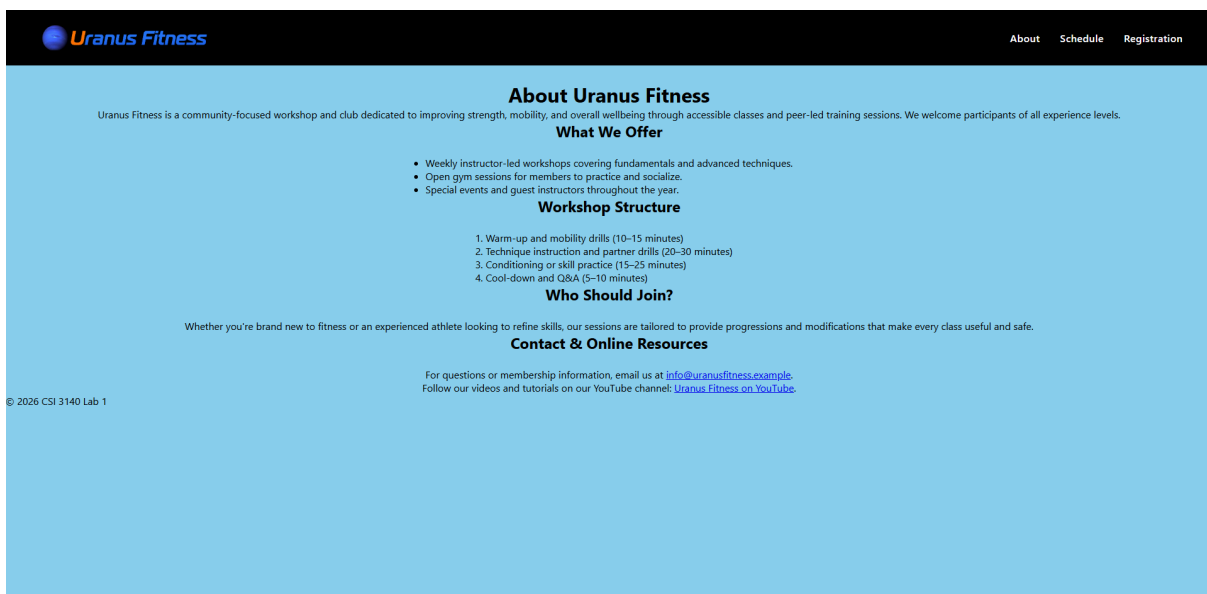
Yes — Chrome/Firefox

8. Screenshots

- index.html in the browser



- about.html in the browser



- schedule.html in the browser


[About](#)
[Schedule](#)
[Registration](#)

Available Plans in Uranus Fitness

Uranus Fitness Plans

Plans	Duration	Type of workout	Price	Select
Beginner Plan	2 weeks	Machine assisted Exercises	\$49	Click Here
Intermediate Plan	5 weeks	Free weight Exercises	\$59	Click Here
Advanced Plan	8 weeks	Compound Lifts + Diet Plans	\$69	Click Here

What will you become after completing our programs

Gym Rat
Gym Bruh
Personal Trainer
Responsible Gym member
Physically fit individual. T&C apply
[www.standards class - lab1 - schedule.html](#)

- registration.html — empty form


[About](#)
[Schedule](#)
[Registration](#)

Registration Form

Personal Information:

Name:

Email:

Age:

Membership Tier:

Select Tier:

☐ Beginner
☐ Intermediate
☐ Advanced

About You

Tell us about yourself:

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<file:///D:/3140/661/3104/Labs/WWW-Structures-and-Standards/Lab1/index.html>

- W3C validation result for each of the four pages

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for about.html (checked with vnu 26.5.29)

Checker Input

Show ☐ source ☐ outline ☐ image report ☐ errors & warnings only [Options...](#)

Check by [file upload](#) [Choose File](#) No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

[Check](#)

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 3 milliseconds.

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Showing results for index.html (checked with vnu 26.5.29)

Checker Input

Show ☐ source ☐ outline ☐ image report ☐ errors & warnings only [Options...](#)

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Document checking completed. No errors or warnings to show.

Used the HTML parser.

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Showing results for registration.html (checked with vnu 26.5.29)

Checker Input

Show ☐ source ☐ outline ☐ image report ☐ errors & warnings only [Options...](#)

Check by file upload ▾ [Choose File](#) No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

[Check](#)

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 6 milliseconds.

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Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for schedule.html (checked with vnu 26.5.29)

Checker Input

Show ☐ source ☐ outline ☐ image report ☐ errors & warnings only [Options...](#)

Check by file upload ▾ [Choose File](#) No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

[Check](#)

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 3 milliseconds.

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9. Reflection

Valid HTML and accessible HTML turned out to be related but not identical. The validator caught structural problems early (a malformed void element, stray closing tag, an obsolete attribute) but stayed silent on issues that mattered just as much: our first form relied on placeholder text instead of real labels; it passed validation but would have failed a screen reader so we added explicit `<label for=>/id` pairs and grouped controls in `<fieldset>`. We also tightened heading structure after not every page used a `<h1>` per page. We tested

navigation, the table, and the form in Chrome and Firefox, confirming required-field validation and keyboard navigation behaved consistently.